



A new forecasting model with wrapper-based feature selection approach using multi-objective optimization technique for chaotic crude oil time series



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ABSTRACT

Forecasting the future price of crude oil, which has an important role in the global economy, is considered as a hot matter for both investment companies and governments. However, forecasting the price of crude oil with high precision is indeed a challenging task because of the nonlinear dynamics of the crude oil time series, including chaotic behavior and inherent fractality. In this study, a new forecasting model based on support vector regression (SVR) with a wrapper-based feature selection approach using multi-objective optimization technique is developed to deal with this challenge. In our model, features based on technical indicators such as simple moving average (SMA), exponential moving average (EMA), and Kaufman's adaptive moving average (KAMA) are utilized. SMA, EMA, and KAMA indicators are obtained from Brent crude oil closing prices under different parameters. The features based on SMA and EMA indicators are formed by changing the period values between 3 and 10. The features based on the KAMA indicator are obtained by changing the efficiency ratio (ER) period value, which is considered as fractality efficiency, between 3 and 10. The features are selected by the wrapper-based approach consisting of multi-objective particle swarm optimization (MOPSO) and radial basis function based SVR (RBF-SVR) techniques considering both the mean absolute percentage error (MAPE) and Theil's U values. The obtained empirical results show that the proposed forecasting model can capture the nonlinear properties of crude oil time series, and that better forecasting performance can be obtained in terms of precision and volatility than the other current forecasting models.

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1. Introduction

Crude oil, one of the most significant forms of energy in the world, plays a substantial role in various production processes throughout the world by meeting approximately two-thirds of global energy demand [1–3]. Therefore, volatility in crude oil prices has an important impact on the global economy [4]. Modeling and forecasting these volatilities that affect social stability, economic development and inhabitants in a country enable governmental policy making, as well as the reduction of financial losses in the industrial field [5]. However, the volatilities in crude oil prices generally exhibit complex [6], nonlinear, non-stationary and long-

term memory characteristics [7], this makes it difficult to forecast the crude oil price with high accuracy. Therefore, in order to forecast the crude oil price with high precision, it is necessary to cope with the nonlinear dynamics [8–10] of the crude oil time series including chaoticity [11] and inherent fractality [12]. In this study, we propose a novel crude oil price forecasting model based on radial basis function based support vector regression (RBF-SVR) with a wrapper-based feature selection approach using meta-heuristic based multi-objective particle swarm optimization (MOPSO) algorithm to cope with the nonlinear dynamics of crude oil price time series. The performance of the proposed forecasting model has been empirically tested on Brent crude oil time series.

The price of crude oil, which has a major impact on the global economy, is determined by many predicted or unpredictable factors, such as organization of petroleum exporting countries (OPEC)

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oil policy, political events, financial speculations and shocks, government interventions, weather conditions, stock market, exchange rates, dollar index, gold, interest rate, supply, demand, unemployment and so on [13,14]. These factors may induce to a largely nonlinear and chaotic tendency of crude oil price time series [15,16]. Moreover, the formation of crude oil prices is determined by the simultaneous interaction between these factors, which causes the crude oil price time series sufficiently complex. This situation shows the existence of multifractality and phase transitions in the crude oil price time series [12]. Therefore, it is difficult to forecast the volatility and trends in crude oil prices with high accuracy and precision.

Traditional econometric techniques, such as autoregressive integrated moving average (ARIMA), generalized autoregressive conditional heteroskedasticity (GARCH) models, linear regression (LR), vector autoregression (VAR), co-integration analysis, grey model, naive random walk, error correction models (ECM), are insufficient to capture these nonlinear dynamics involving inherent fractality and chaoticity in crude oil price time series [17,18]. Instead of these traditional econometric techniques, models such as artificial neural networks (ANNs) and support vector machine (SVM) based on artificial intelligence (AI) models capable of self-learning have been used in the forecasting of crude oil price time series. It has been empirically demonstrated that these models are superior to traditional econometric models in forecasting crude oil price time series [19,20]. However, some problems such as loss of time, slow convergence, local minima and complex algorithm emerge as the lack of AI models. Although these artificial intelligence models outperform traditional econometric models, when used as single model, they are insufficient to forecast volatility in crude oil price time series. In order to overcome these problems and to forecast the trend and volatility of crude oil price time series with high accuracy, hybrid models based on the principle of "decomposition and ensemble" have been used [21–23].

In [24], the forecasting of crude oil price time series has been performed by hybrid grey wave forecasting method, which combines graphical forecasting method with autoregressive moving average (ARMA) or random walk (RW). The forecasting model is formed based on both the information obtained from the data and the graphical information of the time series graph. It is specified that the obtained model performs better in root mean square error (RMSE) and Theil's inequality coefficient (TIC) than traditional econometric models. In Ref. [25], the crude oil price forecasting approach based on stream learning has been proposed to analyze and make inference of continuous flow of non-stationary data. It is emphasized that the proposed model will continually improve over time by updating itself when a new crude oil price data arrives, thus can capture the changing pattern of oil price time series. Chiroma et al. [26] have proposed a new alternative combined model, including the genetic algorithm (GA) and neural network (NN) to forecast the crude oil price. The GA in the model optimizes the weight coefficients, topology, and bias of NN both to improve convergence speed (CS) and to reduce the complexity of the NN structure. In Ref. [17], the crude oil price time series forecasting has been performed by empirical mode decomposition (EMD) based NN ensemble learning model. The proposed model consists of three stages. In the first stage of the model, the original crude oil price time series is decomposed into intrinsic mode functions (IMFs). In the second stage, each of the IMFs is modeled with a three-layer feed-forward NN (FNN). In the final stage, the forecasting results of all IMFs are combined with an adaptive linear NN (ALNN) to formulate an ensemble output for these time series. In Ref. [27], in order to analyze the effects of internet concern (IC) on the oil market and forecast the volatility in crude oil price, the relationship between the IC and the volatility in oil price have been examined

using an extreme learning machine (ELM) model. It has experimentally shown that the ELM models, involving IMFs and IC, perform better than baseline ELM and other benchmark models at different horizons. In Ref. [21], a new decomposition-and-ensemble learning paradigm combining ensemble EMD (EEMD) and extended ELM (EELM) has been proposed for the estimation of the crude oil price time series. In the first phase of the model, the IMF components are decomposed by the EEMD method applied to the crude oil price time series. In the second phase, EELM, which is proposed as effective, powerful and stable prediction tool, is applied to estimate all components independently. In the final phase, the estimated results are added into an ensemble result as final estimation. It has empirically illustrated that the proposed paradigm outperforms benchmark models in terms of both efficiency and estimation accuracy. In Ref. [28], a non-iterative EEMD based random vector functional link (RVFL) network model has been developed for crude oil price forecasting. It is evidenced that the proposed model is significantly superior to all considered comparison methods in terms of robustness, speed and accuracy. In Ref. [29], a random wavelet neural network (RWNN) model has been proposed to predict volatility in crude oil prices. The proposed model has been reported to have better accuracy in crude oil price volatilities forecasting than back propagation neural network (BPNN) and SVM models.

This study is the first study that can predict volatility in crude oil price time series with high accuracy and precision by using RBFsVR with wrapper-based feature selection approach handling multi-objective optimization technique. The main contribution of this study to the literature is that the volatility in crude oil price is estimated by higher performance in terms of both accuracy and robustness compared to other studies in the literature thanks to the combination of feature selection and multi-objective optimization algorithms. Thus, estimates of events that will develop depending on future crude oil prices will be able to be used for national and international development plans by decision makers. The remaining part of this paper is organized as follows. All methodology, including technical indicators, forecasting model with feature selection technique, RBFsVR, and MOPSO algorithm is introduced in Section 2. The empirical results of proposed model for crude oil price forecasting are discussed in Section 3. Finally, the conclusions are presented in Section 4.

2. Methodology

This section describes our approach, which has high performance to forecast volatility in the crude oil price. In our proposed model, the features based on technical indicators such as simple moving average (SMA), exponential moving average (EMA), and Kaufman's adaptive moving average (KAMA) have been extracted. The extracted features are selected by a wrapper-based approach consisting of MOPSO and RBFsVR techniques, taking **mean absolute percentage error** (MAPE) and Theil's U values into consideration. The methodology used in our model is introduced in the following subsections.

2.1. Technical indicators

Technical indicators are frequently used in financial estimates due to their advantages in eliminating inherent noise in time series and demonstrating structural factors affecting change. In this study, the moving averages including SMA and EMA and the adaptive moving average are used as technical indicators to identify the direction of the trend and to take into account the noise and volatility in the time series.

2.1.1. Moving averages

The moving averages (MA) are a good way to measure momentum as well as validate trends and identify support and resistance areas. Also MAs, also known as lagging indicator plotting average asset prices over time, are also used as a smoothing device to verify the emergence of new trends. These indicators, which follow the price by smoothing the price data, are not used alone to estimate the price direction, but are used to determine the current direction of the trend [30]. Shorter-length averages tend to follow changes in underlying asset prices more closely, while longer-length averages only desensitize asset price movements as an alternative, highlighting only major trends [31]. SMA, which is the simple average of a security over a defined number of time periods and EMA, which gives bigger weight to more recent prices exponentially are the most commonly used technical indicators. The SMA can be defined as

$$SMA_n = \frac{1}{n} \sum_{t=k-n+1}^k C_t \quad (1)$$

where n is the number of periods included in the average, k is the relative position of the current period observed, and C_t is the current closing price at time t . The EMA, which is more appropriate than the SMA to determine markets trends, can be calculated as

$$EMA_n = C_{t-1} \times p - EMA_{n-1} \times (p - 1) \quad (2)$$

where C_{t-1} is the closing price in the previous period, n is the number of periods included in the EMA calculation, p is smoothing factor, EMA_n and EMA_{n-1} are the values of EMA in the current and previous period, respectively [31]. The smoothing factor is defined as $p = \frac{2}{n+1}$ and the limiting values are $0 < p < 1$.

2.1.2. Kaufman's adaptive moving average

The adaptive MA, proposed by Kaufman, called as KAMA, is based on the premise that a short-length MA will respond faster when market prices are trending, yet a long-length MA that varies between markets will be preferred [31]. It is designed to use the fastest possible trend based on the smallest calculation period for current market conditions. The KAMA aims to define and adapt to changing market conditions with the help of efficiency ratio (ER). The ER, also known as fractal efficiency, measures the relative speed of market movement from one price level to another. It has the value of 1 when prices move in the same direction for full n periods, and a value of 0 when prices are unchanged over the n periods [32]. The ER should enable the KAMA to produce trade signals rapidly in the market in order to exploit the price trend better, and in a changing market the usual "whipsaw" effect of over trading should be avoided.

The KAMA indicator is represented by three periods and shown as $KAMA(T_1, T_2, T_3)$. Note that T_1 is the period value of the ER, T_2 is the period value of the EMA, and T_3 is the period value of the slow EMA. The KAMA can be described as

$$KAMA_t = KAMA_{t-1} + S_t \times (C_t - KAMA_{t-1}) \quad (3)$$

where C_t is the current closing price, $KAMA_t$ is the current adaptive MA value, $KAMA_{t-1}$ is the previous adaptive MA value and S_t is the smoothing coefficient at time t [31]. The S_t is calculated each period as

$$S_t = [ER_t \times (fastest - slowest) + slowest]^2 \quad (4)$$

where

$$fastest = 2 / (fastest\ MA\ period + 1) \quad (5)$$

$$slowest = 2 / (slowest\ MA\ period + 1) \quad (6)$$

$$ER_t = \frac{|C_t - C_{t-n}|}{\sum_{i=t-n+1}^t |C_t - C_{t-i}|} \quad (7)$$

2.2. Forecasting model with feature selection technique

Feature selection is defined as the selection of the best subset that can represent the original data set. According to the algorithm used, each potential feature is evaluated by scoring according to a particular feature selection metric and then the best k features are selected among the n features in the data set [33]. By reducing the number of features, noisy data and unrelated data are eliminated, and as the size of the feature set is reduced, the algorithm is enabled to run quickly. The methods used in feature selection are generally divided into three groups: filtering methods based on statistical information, wrapper methods that perform search operations on features, and embedded methods based on interaction with the classification model [34]. In this study, the most useful and most important features for our crude oil price forecasting model are selected by wrapper method and the number of features in the data set is reduced. The wrapper method is used because of its high performance in determining the best sub-feature set.

2.2.1. Radial basis function based support vector regression

Support Vector Machine (SVM), a powerful data mining tool used to solve classification problems, is defined as a vector space-based machine learning method that finds a decision boundary between two classes farthest away from any point in training data [35]. The purpose of the SVM is to determine the hyperplane that will optimally separate the classes of the target variable from each other. The SVM ensures a general mechanism for fitting the hyperplane surface to the training data with the help of a kernel function such as polynomial, linear, radial basis, or sigmoid. The SVM can also be applied to nonlinear data as well as linear data. The fact that the data is in a structure which can be separated linearly or is in a structure which cannot be separated linearly emerges as two situations that can be encountered in SVMs. How to set the best kernel parameters and how to select the most appropriate subset of input features for the SVM are two important problems in using the SVM. Since the selection of the feature subset affects the appropriate kernel parameters, the classifier's performance will also be directly affected from this situation.

Support Vector Regression (SVR) is presented as an alternative to the ϵ -insensitive loss function, which uses structural reduction principles to select discriminative functions with minimum risk limitations [35,36]. The nonlinear SVR tries to find a regression function in the hyperspace expressed by

$$f(x) = w^T \phi(x) + \beta \quad (8)$$

where $w = [w_1, \dots, w_m]$ is the weight vector, $\phi(x) = [\phi_1(x), \dots, \phi_m(x)]$ is the nonlinear mapping function, and β is the bias. The function that maps the nonlinear x data into a higher, possibly infinitely dimensional feature space is defined as the mapping function and is represented by $\phi(x)$. Suppose we have a set of training data, $\{(x_j, y_j), j = 1, \dots, m\}$ such that $x_j \in \mathcal{R}^n$ and $y_j \in \mathcal{R}$. Note that x_j is the input vector and y_j is the corresponding output, and m indicates the number of data in the training set. The optimal

regression function is obtained by the minimum of the functional

$$R(w, \beta, \xi, \xi^*) = \frac{1}{2} \|w\|^2 + \mu \sum_{j=1}^m (\eta_j(\xi_j) + \eta_j(\xi_j^*)) \quad (9)$$

where μ is the pre-determined parameter, ξ_j is slack variable that indicates whether samples have entered the ε tube, and η_j is loss function [36]. The nonlinear SVR can be obtained by solving the following quadratic programming problem [37]:

$$\begin{aligned} \min R(w, \beta, \xi, \xi^*) &= \frac{1}{2} \|w\|^2 + \mu \sum_{j=1}^m (\eta_j(\xi_j) + \eta_j(\xi_j^*)) (w^T \phi(x)) \\ &\quad + \beta - y_j \\ &\leq \xi_j + \varepsilon \end{aligned} \quad (10)$$

subject to

$$y_j - (w^T \phi(x)) + \beta - y_j \leq \xi_j + \varepsilon \xi_j^*, \quad \xi_j^* \leq 0 \quad (11)$$

Eq. (9) is minimized using Lagrange multiplier techniques. Thus, the optimization problem for SVR changes to

$$\begin{aligned} \max Q(\alpha, \alpha^*) &= \sum_{j=1}^m (\alpha_j - \alpha_j^*) y_j - \frac{1}{2\mu} \sum_{j=1}^m \alpha_j^2 - \alpha_j^{*2} + \frac{1}{2} \sum_{j=1}^m \\ &\quad \times \sum_{l=1}^m (\alpha_j - \alpha_j^*) (\alpha_l - \alpha_l^*) K(x_j, x_l) \end{aligned} \quad (12)$$

subject to the constraint

$$\sum_{j=1}^m (\alpha_j - \alpha_j^*) = 0, \quad 0 < \alpha_j, \alpha_j^* < \mu \quad (13)$$

The regression function with kernel function can be defined as

$$f(x, \alpha) = \sum_{j=1}^m (\alpha_j^* - \alpha_j) K(x, x_j) + \beta \quad (14)$$

where α and α^* are Lagrange multipliers, and $K(x, x_j)$ is a kernel function [36]. In this case, the nonlinear radial basis function (RBF) kernel is described as

$$K(x, x_j) = \exp(-\gamma \|x - x_j\|^2) \quad (15)$$

where γ is the parameter of the kernel function. The different kernel functions within the SVR offer different approach capabilities. The kernel values in the SVR have a significant effect on forecast performance [38]. The Gaussian RBF is adopted as SVR kernel in this study.

2.2.2. Multi-objective particle swarm optimization

Evolutionary algorithms can be defined as a subset of evolutionary computations that can produce optimal solutions for linear and nonlinear problems in continuous or discrete time using

validated experimental theories of biological evolution. These algorithms are often used in multi-objective optimization problems since no technique is used, such as the weighting approach directly related to the opinions of decision-makers, and a set of solutions can be obtained from a program run [39]. Evolutionary algorithms are often preferred to solve problems in many different fields [39,40]. Particle Swarm Optimization (PSO) is a swarm intelligence algorithm based on birds' social interactions and field of evolutionary computing, inspired by birds acting as a swarm [41]. In the PSO algorithm, each bird is referred by a particle and a set of these particles is defined as a swarm. Each particle adjusts its position to the best position in the swarm, using its previous experiences. The PSO is mainly based on bringing the position of individuals in the swarm near to the individual with the best position of the swarm. This approach speed is a situation that develops randomly, and most of the time the individuals in the swarm are better in their new movements than in the previous positions, and this process continues until it reaches the target.

Consider the following definitions. The position, velocity and individual experience of the p th particle are defined as respectively,

$$X_p = [X_{p,1}, X_{p,2}, \dots, X_{p,d}] \quad (16)$$

$$V_p = [V_{p,1}, V_{p,2}, \dots, V_{p,d}] \quad (17)$$

$$P_p = [P_{p,1}, P_{p,2}, \dots, P_{p,d}] \quad (18)$$

where S is swarm size, d is the number of decision variables, and $1 \leq p \leq S$.

The position and velocity attributes of the particle are updated and the behavior of the particle towards the optimal solution is managed. The velocity and position of each particle are updated in step $p + 1$ as

$$V_{p,r} = \omega V_{p,r} + c_1 R_1 (Pbest_{p,r} - X_{p,r}) + c_2 R_2 (Gbest_{p,r} - X_{p,r}) \quad (19)$$

$$X_{p,r} = X_{p,r} + V_{p,r} \quad (20)$$

where $c_1, c_2, \omega \geq 0$, and $1 \leq r \leq d$. c_1 and c_2 are acceleration coefficients, R_1 and R_2 are random numbers, which are generated uniformly in the range $[0, 1]$, and ω is the inertia weight. The terms $c_1 R_1 (Pbest_{p,r} - X_{p,r})$ and $c_2 R_2 (Gbest_{p,r} - X_{p,r})$ are referred to as cognitive term and social term, respectively [42]. The individual experience of the particle is stated in the cognitive term. The interaction between the particles is expressed in the social term. For the d [43]th dimension in the $(p + 1)$ th iteration, $Pbest_{p,r}$ is the best position of the p th particle for, and $Gbest_{p,r}$ is the best position of the swarm. The variation of the velocity of a particle is controlled by the lower and upper bounds V_{min} and V_{max} limit values in Eq. (21) specified by the user. The position of particle is computed by using Eq. (20).

$$V_{min} \leq V_{p,r} \leq V_{max} \quad (21)$$

For the simulation model in the next step, the position of the particle is used as new input and new objective functions are

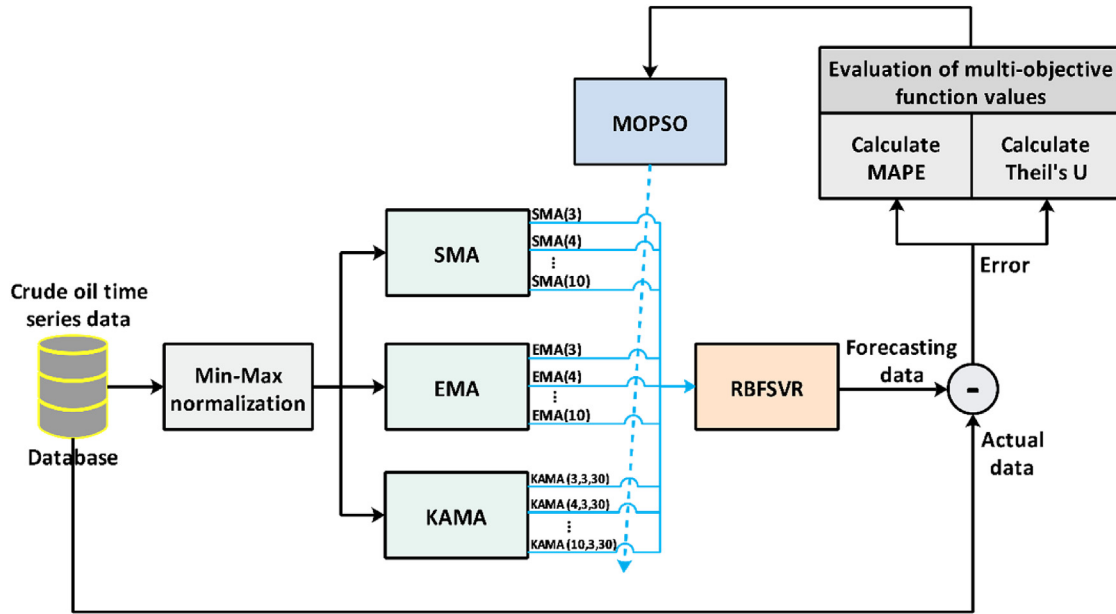


Fig. 1. Structure of proposed crude oil price forecasting model.

Table 1
Error metrics used in the evaluation of our model.

Metric	Mathematical Expression
MAPE	$MAPE = \frac{1}{N} \sum_{h=1}^N \left \frac{F_{forecasted}^h - F_{actual}^h}{F_{actual}^h} \right \times 100\%$
Theil's U	$Theil's U = \frac{\sqrt{\frac{1}{N} \sum_{h=1}^N (F_{actual}^h - F_{forecasted}^h)^2}}{\sqrt{\frac{1}{N} \sum_{h=1}^N (F_{actual}^h)^2} + \sqrt{\frac{1}{N} \sum_{h=1}^N (F_{forecasted}^h)^2}}$

computed. Each particle follows the objective function in the search process. The process is stopped when the maximum number of iterations is reached. In the MOPSO algorithm, the number of objective functions is more than one and the movement in the decision space towards each objective and therefore the search process should be adopted. In this study, the optimal solution, where both the MAPE value and the standard deviation of the error is minimum, is achieved in MOPSO by two different objective functions.

2.3. Framework of proposed crude oil time series forecasting model

In this study, the proposed model consists of data preprocessing step, feature extraction step and selection of features by wrapper-based approach consisting of MOPSO and RBFSVR algorithms considering both MAPE and Theil's U values. The structure of the proposed crude oil time series-forecasting model is presented in Fig. 1. In the data-preprocessing step, the crude oil time series data are normalized by *min-max* normalization technique. In the feature extraction step, twenty-four features are extracted using attributes based on technical indicators such as SMA, EMA and KAMA. In the crude oil forecasting model identification step, the features are selected by the wrapper-based approach consisting MOPSO and RBFSVR algorithms considering both MAPE and Theil's U values. The optimum solution, where both the MAPE value and the

standard deviation of the error is minimum, is achieved by two different objective functions involving MAPE and Theil's U. The performance of our model has been measured by taking MAPE and

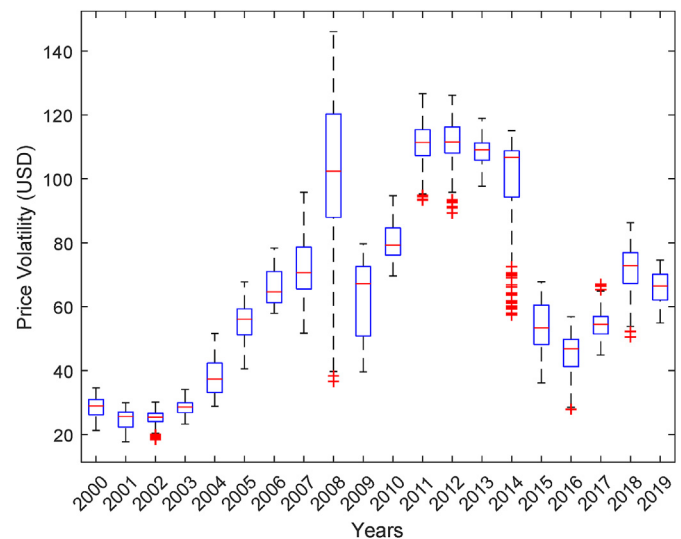


Fig. 2. Crude oil closing price volatility versus years.

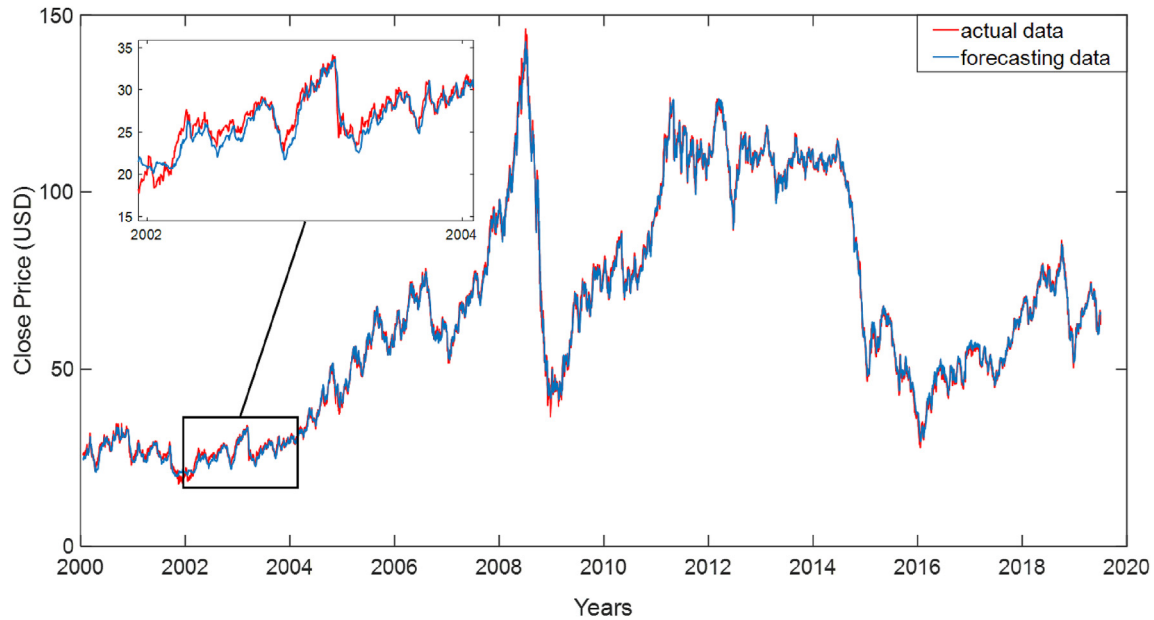


Fig. 3. Forecasting and actual values of crude oil closing price time series for RBFSVR-PSO hybrid model.

Theil's U into consideration. The mathematical expression of these error metrics is presented in Table 1 [43]. The accuracy and volatility of our model have been verified by these indexes.

3. Empirical results and discussion

In this section, the studies carried out using Brent crude oil price data are presented to prove the forecasting performance of our model. Besides, the effect of MOPSO algorithm on crude oil price forecasting model has been examined. In this study, briefly, one-step ahead forecasting price of crude oil is obtained with the RBFSVR model using Brent crude oil data. The data set containing daily closing prices in US dollars for Brent crude oil is used. The time

series, which involves daily closing prices in US dollars, includes data from 04 January 2000 to 02 July 2019 for Brent crude oil. The crude oil time series used in the study consists of 5000 samples. The minimum, maximum, mean and standard deviation values of the time series used in the study are 17.68, 146.08, 65.3223, and 30.3408, respectively.

The volatility of the crude oil closing price versus the years is presented in Fig. 2. The price volatility of crude oil varies at different time intervals. For example, the Brent crude oil prices, which ranged between 20 USD and 30 USD from 2000 to 2003, showed a rising trend ranging between 35 USD and 140 USD from 2003 to 2008. In order to define a generalized forecasting model that takes these differences into account, 10-fold cross validation method has

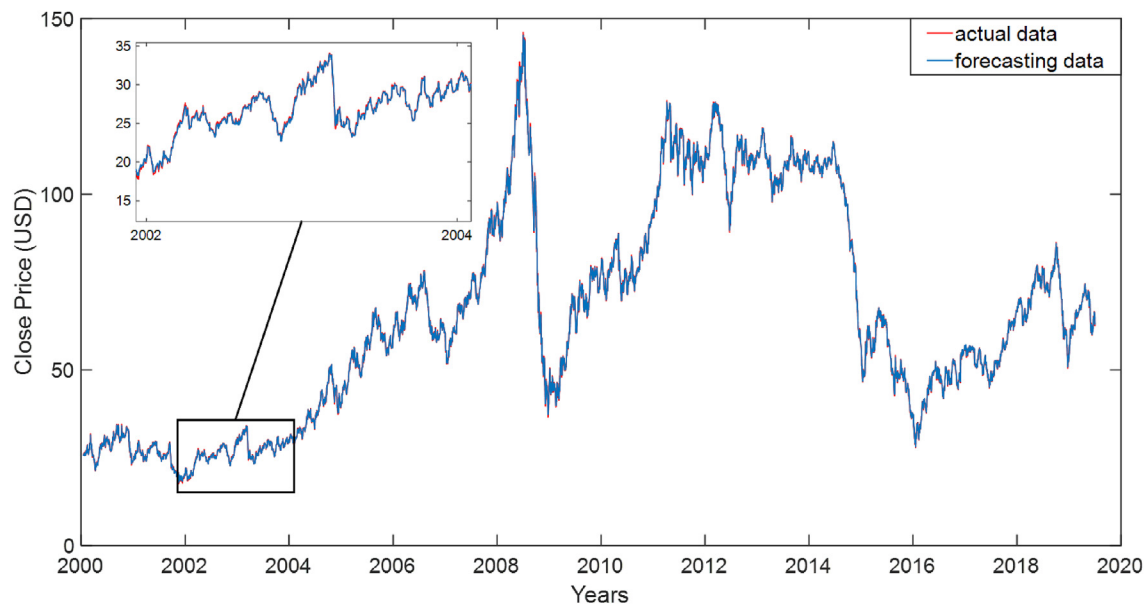


Fig. 4. Forecasting and actual values of crude oil closing price time series for RBFSVR-MOPSO hybrid model.

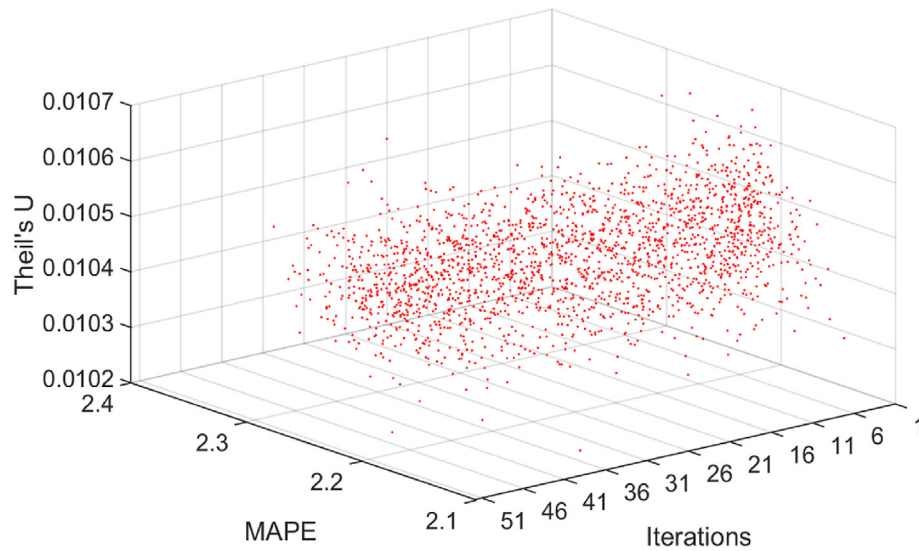


Fig. 5. Changes of MAPE values versus Theil's U during each iteration for PSO.

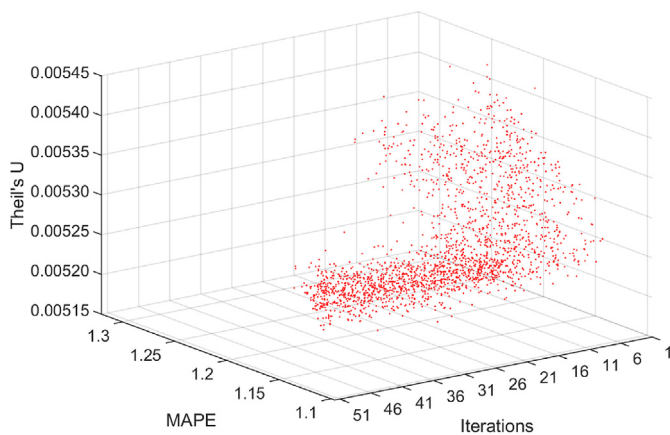


Fig. 6. Changes of MAPE values versus Theil's U during each iteration for MOPSO.

been used in the study. Indicator arrays have been formed for different parameter values including features based on SMA, EMA and KAMA indicators. The features based on SMA and EMA indicators have been formed by changing the period values between 3 and 10. The features based on the KAMA indicator have been obtained by changing the ER period value, which is considered as fractality efficiency, between 3 and 10. Both PSO and MOPSO methods are used to select the indicators that can improve the model performance from these indicator arrays. Two forecasting models have been identified by the wrapper-based approach consisting both optimization methods and RBFSVR algorithms. The forecasting performances of the crude oil closing price time series have been determined for two defined models consisting of PSO and MOPSO and the obtained results are presented in Figs. 3 and 4, respectively. It is seen that the model formed with the features selected for MOPSO has higher forecasting performance.

Multi-objective optimization in the model is performed by taking MAPE and Theil's U values into account. The changes of the

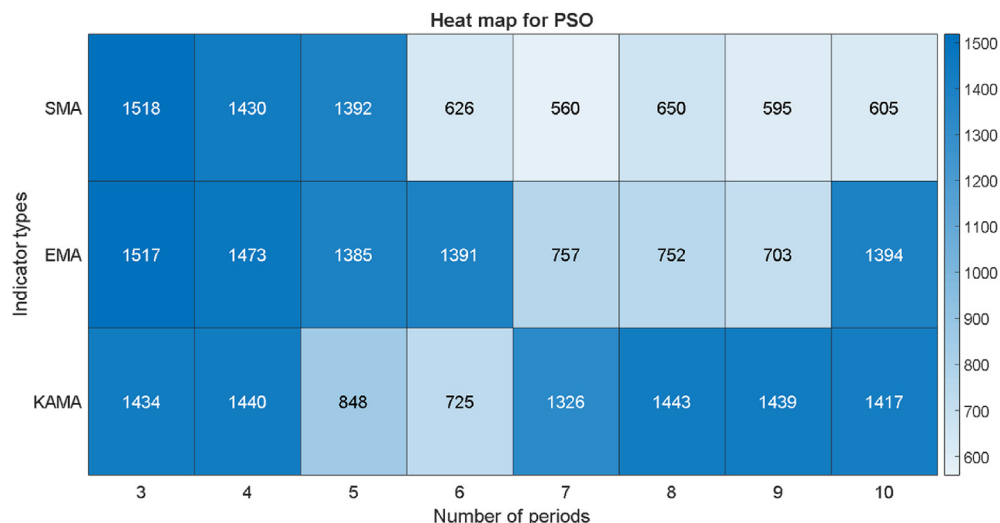


Fig. 7. The most selected features for PSO in all iterations.

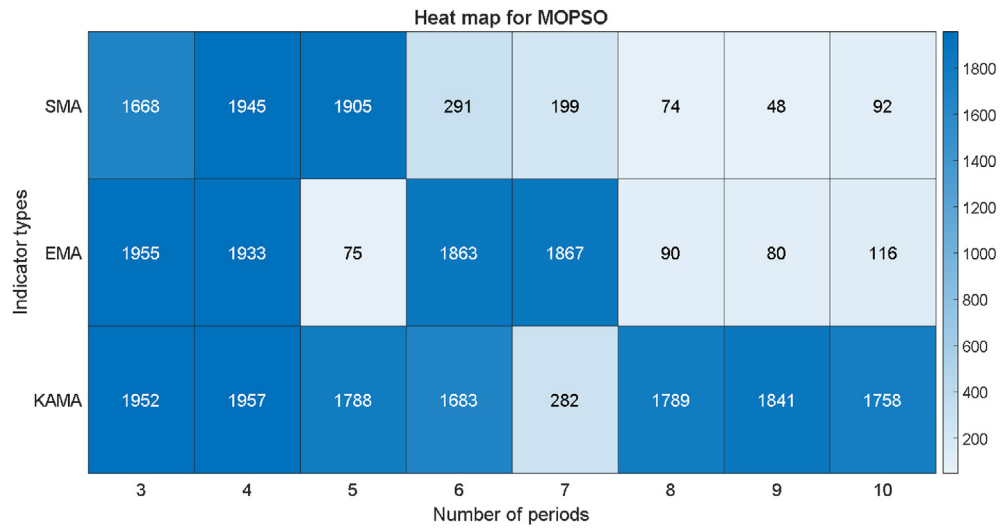


Fig. 8. The most selected features for MOPSO in all iterations.

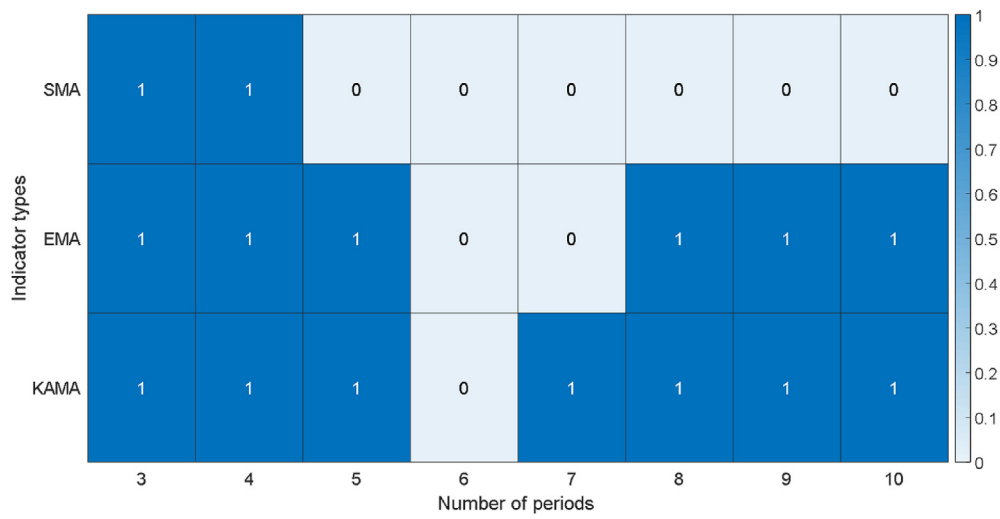


Fig. 9. Selected features for the model that gives the least MAPE value in the PSO.

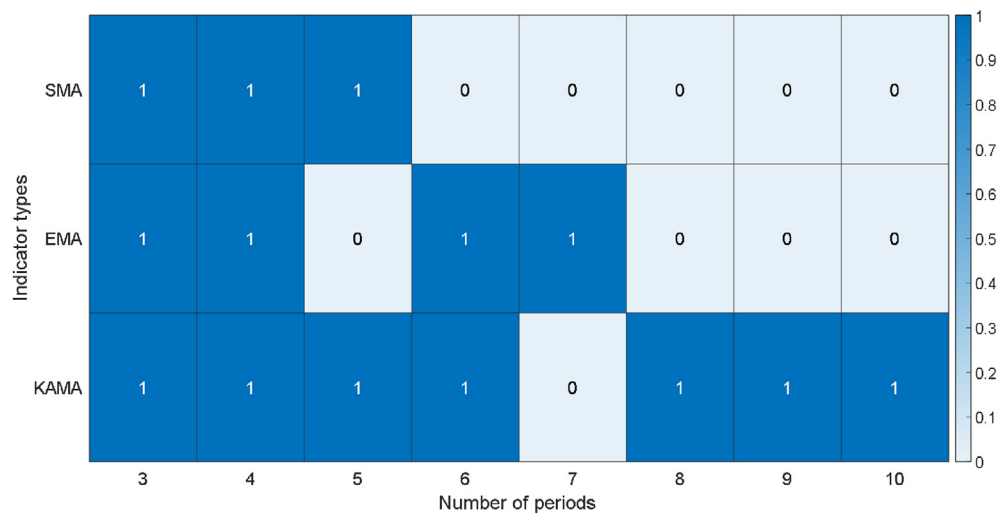
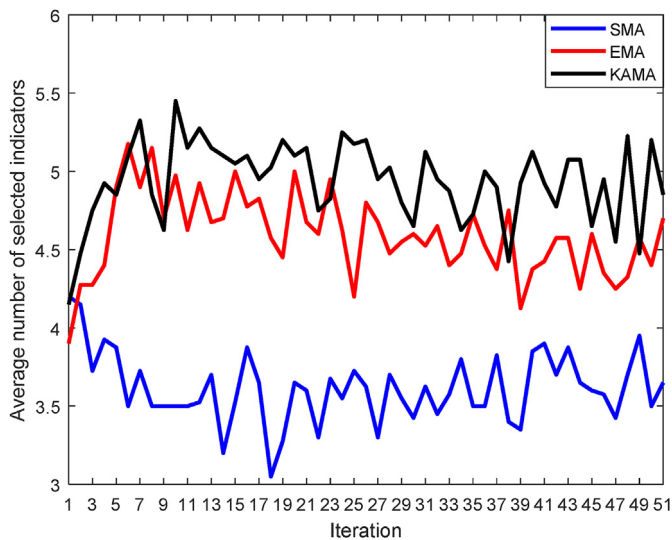
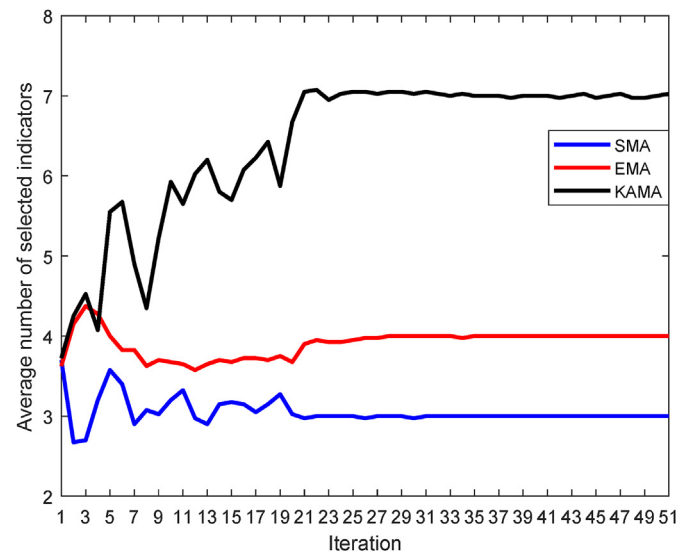


Fig. 10. Selected features for the model that gives the least MAPE value in the MOPSO.

Table 2

The parameter values of the algorithms used in the study.

	Parameter	Attribute	Value
PSO	S	number of particles	40
	d	maximum number of iterations	50
	c_1	cognitive factor	2
	c_2	social factor	2
	V_{max}	maximum velocity	6
	ω_{max}	maximum bound on inertia weight	0.9
MOPSO	ω_{min}	minimum bound on inertia weight	0.4
	S	number of particles	40
	d	maximum number of iterations	50
	Rep	repository size	100
	ω	inertia weight	0.5
	ω_{damp}	inertia weight damping rate	0.99
	c_{p1}	personal learning	1
	c_{p2}	global learning coefficient	2
	n_{grid}	number of grids per dimension	7
	α	inflation rate	0.1
	β	leader selection pressure	2
	γ	deletion selection pressure	2
	m_u	mutation rate	0.1
	$box\ constraint$		Response Scale/1.35
RBFSVR	ϵ		Response Scale/13.5
		kernel function	Radial Basis Function
		kernel scale	1.2

**Fig. 11.** Average number of selected features for SMA, EMA, and KAMA indicators during each iteration in the PSO.**Fig. 12.** Average number of selected features for SMA, EMA, and KAMA indicators during each iteration in the MOPSO.

values of these two objective functions for possible solutions during each iteration are shown in Figs. 5 and 6 for PSO and MOPSO, respectively. It is noteworthy that in the PSO method, error indicators are integrated into only one objective function. As iteration number progresses, it is seen that the objective functions take values in a wide range. In MOPSO, as the iteration number progresses, the spectrum of the values taken by the objective function narrows since the solutions on the pareto optimal surface are taken into account. It can be said that these solutions provide better forecasting performance when compared to the PSO.

The most selected features for PSO in all iterations are given in Fig. 7. The rates of selection of SMA, EMA and KAMA indicators during all iterations by PSO method are determined as 27.50%, 34.94%, and 37.56%, respectively. The most selected step numbers are 3, 4 and 5 for the SMA indicator, 3, 4, 5, 6 and 10 for the EMA indicator, and all step numbers except 5 and 6 for the KAMA

indicator. The most selected features for MOPSO in all iterations are shown in Fig. 8. The rates of selection of SMA, EMA and KAMA indicators during all iterations by MOPSO method are determined as 22.83%, 29.28%, and 47.89%, respectively. The most selected step numbers are 3, 4 and 5 for the SMA indicator, 3, 4, 6 and 7 for the EMA indicator, and all step numbers except 7 for the KAMA indicator.

The selected features for the model that give the least MAPE value in the PSO are illustrated in Fig. 9. The total of 15 features has been selected as 2, 6 and 7 for SMA, EMA and KAMA, respectively. The selected features for the model that give the least MAPE value in the MOPSO are demonstrated in Fig. 10. The total of 14 features has been selected as 3, 4 and 7 for SMA, EMA and KAMA, respectively. When MOPSO and PSO are compared considering the number of features selected, it can be said that the model with MOPSO has less model complexity than PSO. The parameter values

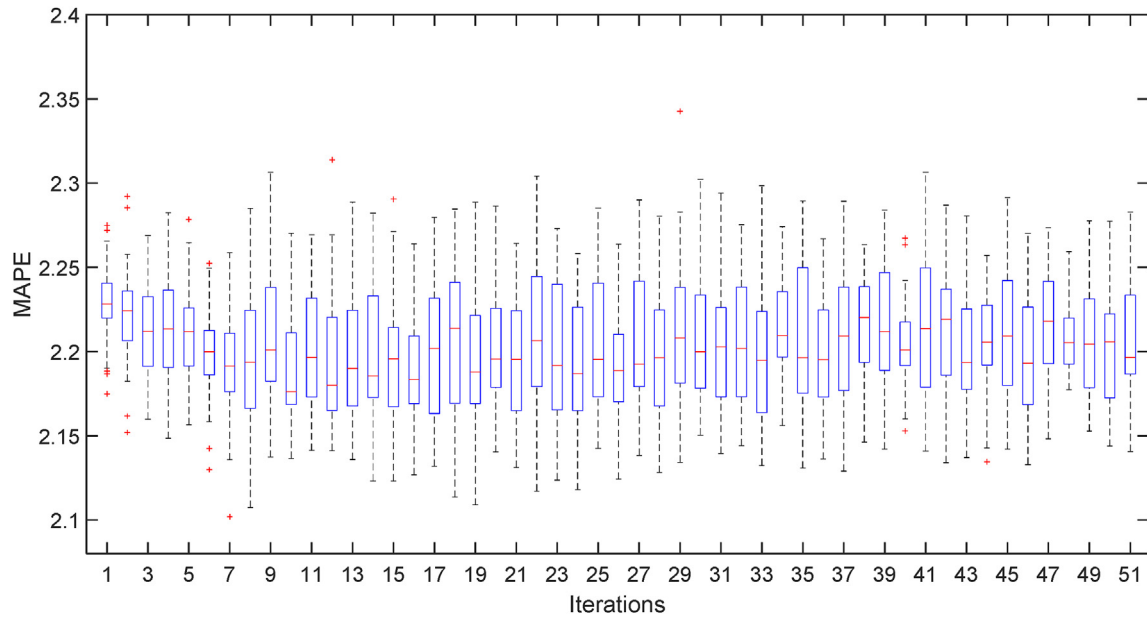


Fig. 13. Changes of MAPE values during each iteration for PSO.

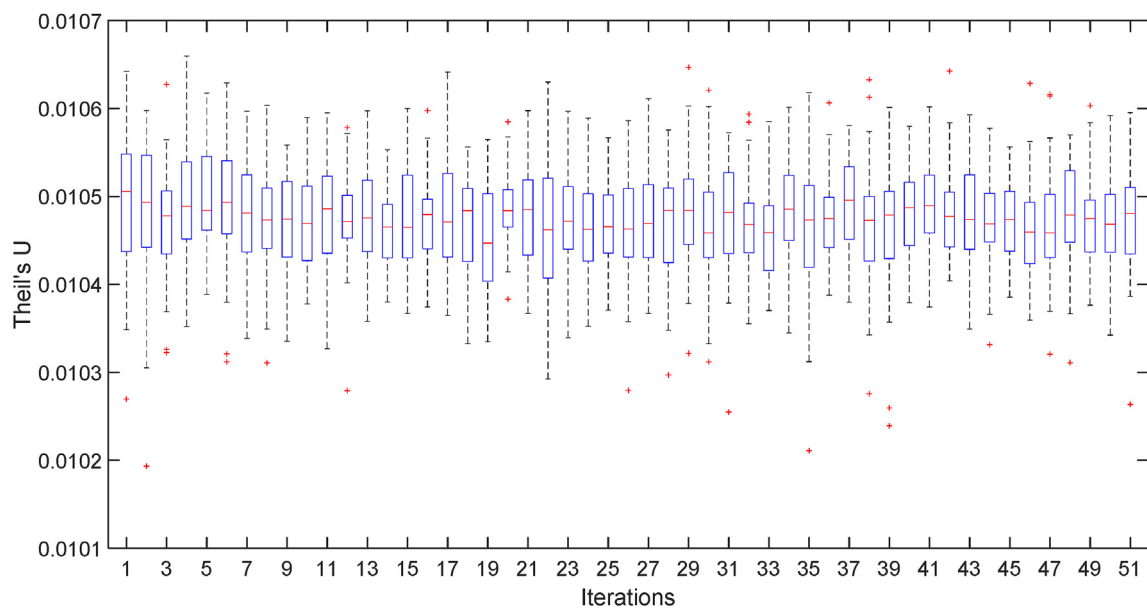


Fig. 14. Changes of Theil's U values during each iteration for PSO.

of the algorithms are used in the study. The parameter values of the algorithms including PSO, MOPSO, and RBF-SVR used in the study are presented in Table 2.

The selected average feature numbers for the SMA, EMA, and KAMA indicators during each iteration in PSO and MOPSO are shown in Figs. 11 and 12, respectively. It is seen that volatility decreases for all indicators in MOPSO while the average number of selected indicators during the iteration is volatile for all indicators in the PSO. The average SMA, EMA and KAMA indicators are 3.65, 4.70, and 4.85 for PSO and 3, 4, and 7.02 for MOPSO, respectively. When Figs. 3–12 are examined together, it is seen that KAMA indicators are selected more with MOPSO and the model which has the best forecasting performance is formed with MOPSO for crude oil.

In our model, the features based on the technical indicators have been used and 24 features are obtained. The features have been selected by the wrapper-based approach consisting of MOPSO and RBF-SVR techniques taking both MAPE and Theil's U values into account. The effect of MOPSO on model performance has been specified by comparing it with PSO, which has a single objective function. The changes in MAPE and Theil's U values during each iteration for PSO and MOPSO are shown in Figs. 13–16. When the box whisker representations in MAPE and Theil's U changes are examined, it is seen that the MAPE and Theil's U values of MOPSO are lower than the MAPE and Theil's U values of PSO during iteration. The PSO and MOPSO approaches have been run for 50 iterations with 40 possible solutions. After 50 iterations MAPE and Theil's U values are obtained as 2.14075 and 0.010418 for the best

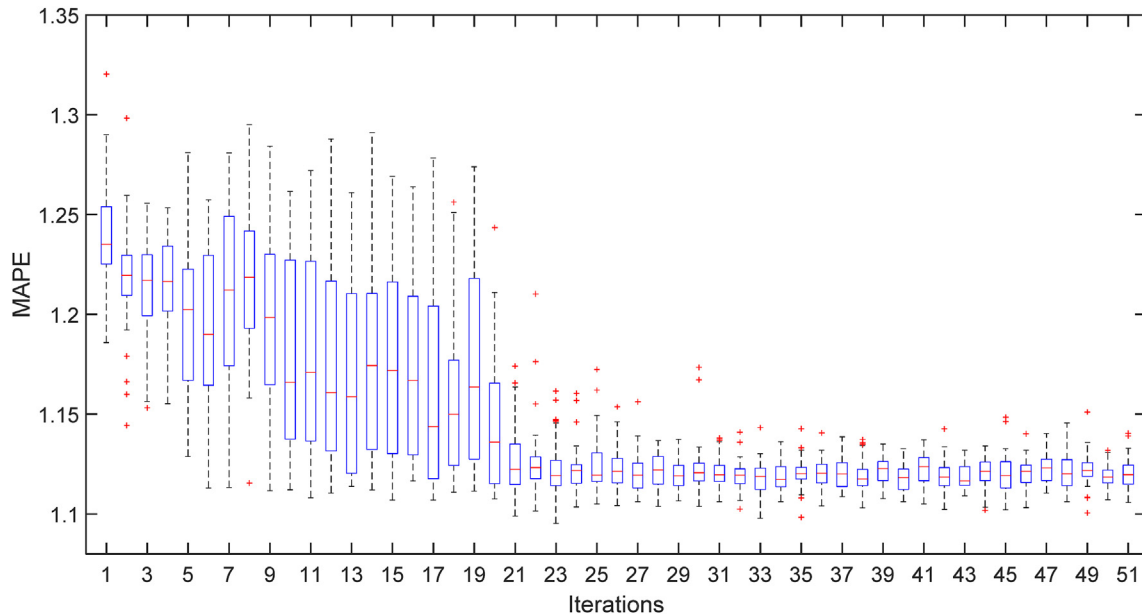


Fig. 15. Changes of MAPE values during each iteration for MOPSO.

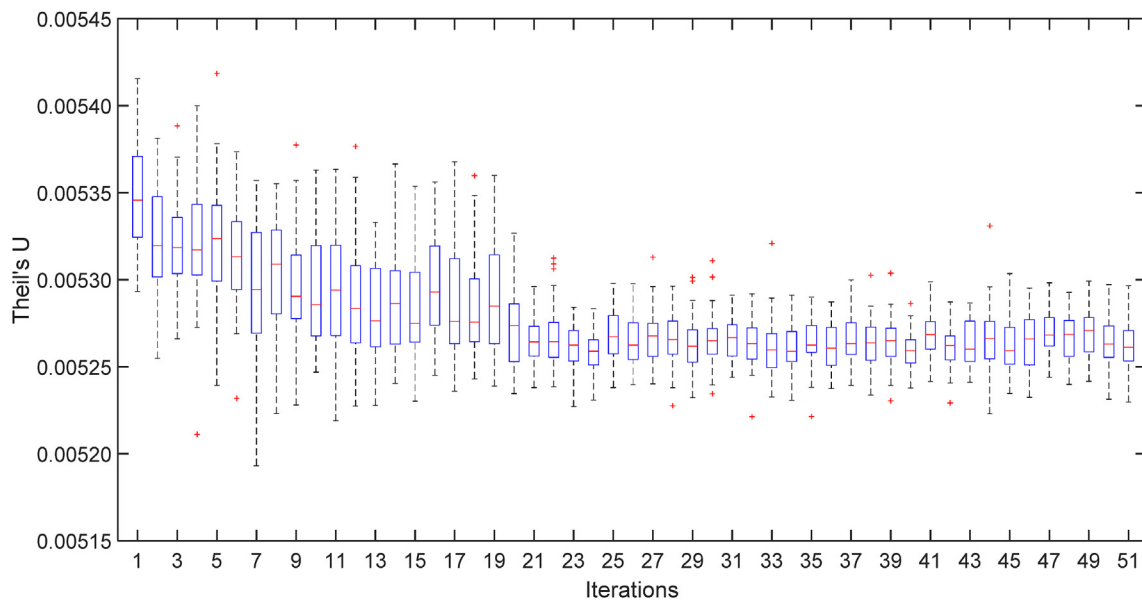


Fig. 16. Changes of Theil's U values during each iteration for MOPSO.

model with PSO and 1.10580 and 0.005239 for the best model with MOPSO, respectively.

4. Conclusions

Forecasting the future price of crude oil, which is one of the most significant forms of energy that cannot be renewed and dispensed within the world, with high accuracy and precision considering the non-linear and time-varying properties in its price time series, is vital for investment companies, financiers, enterprises and governments. Since the volatility in crude oil prices have non-stationary and nonlinear dynamics, which involves chaoticity and inherent fractality, combined models have been proposed using multi-objective optimization technique for crude oil time series

forecasting with high accuracy and precision. In this paper, we have proposed a novel forecasting model based on RBFSVR with a wrapper-based feature selection approach using meta-heuristic based multi-objective optimization technique to deal with this non-stationary and nonlinear dynamics. This study is the first study that can forecast volatility in crude oil time series with high accuracy and precision by using RBFSVR with wrapper-based feature selection approach handling multi-objective optimization technique. In the proposed model, features based on technical indicators such as SMA, EMA, and KAMA have been used and twenty-four features have been obtained. The features have been selected by the wrapper-based approach consisting of MOPSO and RBFSVR techniques considering both MAPE and Theil's U values. The effect of MOPSO on model performance has been determined by

comparing it with PSO, which has a single objective function. The obtained experimental results demonstrate that the proposed model for crude oil time series forecasting can capture the nonlinear properties and that better forecasting performance can be obtained in terms of accuracy and precision than other current forecasting models. In order to test the universality of the proposed model, it can be extended to different real world problems as well as forecasting the future price of crude oil. Future research topics will include improving the proposed methodology for multi-step forecasting of the volatility of crude oil price as well as multivariate analyzes involving several parameters.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2020.118750>.

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